Selected longevity-associated orthologs in major model organisms

Primary lifespan experiments; explicit OMA relationships; illustrative, not an exhaustive survey

	Budding yeast	Nematode	Fruit fly	Mouse
TOR 6 OMA pairs	**TOR1** Deletion **Replicative life ↑** PMID 16293764	**let-363** RNAi **Adult life ↑** PMID 14668850	**Tor** Dominant-negative **Adult life ↑** PMID 15186745	**Mtor** Hypomorph **Median survival ↑** PMID 23994476
S6 kinase 2 OMA pairs	Not selected	**rsks-1** RNAi **Mean life ↑** PMID 17266680	**S6k** Dominant-negative **Adult life ↑** PMID 15186745	**Rps6kb1** Global knockout **Female life ↑*** PMID 19797661
Insulin/IGF receptor 2 OMA pairs	Not selected	**daf-2** Hypomorph **Adult life ↑** PMID 8247153	**InR** Hypomorph **Female life ↑** PMID 11292875	**Igf1r** Heterozygous KO **Female life ↑*†** PMID 12483226

Direct OMA edges: TOR: all six pairs (yeast edges 1:many); S6K: worm–fly, fly–mouse; IIS: worm–fly, worm–mouse (1:many). No fly–mouse edge in this selection.

* Female-specific mouse benefit; corresponding male increase was not significant.

† The original mouse Igf1r increase was not reproduced in a later cohort (PMID 22132081).

TOR and S6K share a pathway; endpoints differ. Blank cells mean not selected, not no effect.

Igf1r is one of two mouse daf-2 receptor paralogs. Sources: GenAge Build 21, OMA All.May2026, cited papers.